(1.) Course Title: INFPHD608 The teaching methodology of Information Technology (complex exam preparatory)	**Credits: 6**
Course **Classification: Mandatory chosen**	
Theoretical or Practical Nature of the Course, *'training character'*[13]: 60/40 (credit%)	
Type of Class: practice and **number of hours**: **2/week** in the given semester, *(If the course is not (only) taught in Hungarian, then **the language**: **English**)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: guided literature review, curriculum document analysis, student presentations, discussion seminar, consultation related to individual research	
Assessment Method (Exam / Midterm Grade / **Other**[3]): practical course mark **Additional** (specific) **methods**[4] applied in knowledge assessment *(if any)*: preparation and oral presentation of an individual, research-based paper (10–12 pages); curriculum analysis and critical reflection	
Course Placement in Curriculum (Semester Number): 4	
Prerequisites (*if any*): -	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

Themes, Possible Questions and related literature
Learning and Teaching:
Questions: What are the strategies that support motivation, the learning process itself, retenion and assessment of learning? How can technology help in this process?
Emergence of Computational Thinking:
Questions: What are emergent computational practices and how can thinking perspectives be improved and assessed?
ICT Competences and algorithmic Skills:
Questions: What is the difference between Computational Thinking and Algorithmic Thinging? What are the methods to improve Algorithmic thinking? What does research say on the levels of Algorithmic Thinking of university students entering Computer Science studies?

2-5 Most Important Required and Recommended Literature
List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

- National Academies of Sciences, Engineering, and Medicine. 2018. How People Learn II: Learners, Contexts, and Cultures. Washington, DC: The National Academies Press. https://doi.org/10.17226/24783. Free: https://www.nap.edu/catalog/24783/how-people-learn-ii-learners-contexts-andcultures
- Kalas I., Blaho A., Moravcik M. (2018) Exploring Control in Early Computing Education. In: Pozdniakov S., Dagienė V. (eds) Informatics in Schools. Fundamentals of Computer Science and Software Engineering. ISSEP 2018. Lecture Notes in Computer Science, vol 11169. Springer, Cham https://link.springer.com/chapter/10.1007%2F978-3-030-02750-6_1

- Brennan, K., Resnick, M.: New frameworks for studying and assessing the development of computational thinking. In: Proceedings of the 2012 Annual Meeting of the American Educational Research Association, Vancouver, Canada, (2012) https://web.media.mit.edu/~kbrennan/files/Brennan_Resnick_AERA2012_CT.pdf
- Francisco Buitrago Flórez, Rubby Casallas, Marcela Hernández, Alejandro Reyes, Silvia Restrepo, Giovanna Danies.: Changing a Generation's Way of Thinking: Teaching Computational Thinking
- László Zsakó, Péter Szlávi (2012): ICT Competences: Algorithmic Thinking. http://dppd.ubbcluj.ro/adn/article_5_2_6.pdf
- Bernát Péter, Zsakó László: Methods of teaching programming – strategy (2017), In: Szerk.: Veronika Stoffová, Szerk.: Roman Horváth New methods and technologies in education and practice: Proceedings of XXX. DIDMATTECH 2017.
  Trnava: Trnava University in Trnava Faculty of Education, 2017. pp. 40-50.
  http://didmattech.truni.sk/2017/proceedings.pdf
- Mária CSERNOCH, Piroska BIRÓ, János MÁTH, Kálmán ABARI : Testing Algorithmic Skills in Traditional and Non-Traditional Programming Environments Informatics in Education, 2015, Vol. 14, No. 2, 175–197, 2015 Vilnius University, DOI: 10.15388/infedu.2015.11
  https://files.eric.ed.gov/fulltext/EJ1079009.pdf
  Through Programming, Volume: 87 issue: 4, page(s): 834-860 Article first published online: May 23, 2017; Issue published: August 1, 2017
  https://journals.sagepub.com/doi/full/10.3102/0034654317710096

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.[1]
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.

[1]**Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.
[2] e.g. case studies, role play, thematic presentations, etc.
[3] e.g. continuous debriefing, mid-year report
[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.
- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.
- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutions.
- Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation.
- Engages as an equal discussion partner with professionals in the field.
- Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.

Course responsible (name, position, academic degree): Dr. Andor Abonyi-Tóth, associate professor

Instructor(s) involved in teaching the course (if any) (name, position, academic degree):
Dr. Andor Abonyi-Tóth, associate professor, PhD., Dr. Győző Horváth, associate professor, PhD.